

Project Proposal

Multi-Agent RAG System for Generating and Evaluating Investment Ideas from Financial News

1. Introduction & Research Objectives

1.1 Research Question

The primary objective of this project is to determine whether a **Multi-Agent RAG (Retrieval-Augmented Generation)** framework can extract actionable investment signals from unstructured financial news that demonstrate a statistically significant predictive value when benchmarked against market data. Specifically, can specialized agents resolve the "noise" in high-volume news to generate hypotheses that outperform a simple sentiment-based baseline?

1.2 Problem Statement

Financial news contains critical market signals, but the volume is too high for manual synthesis, and the data remains largely unstructured. Traditional NLP approaches often fail because:

- **Contextual Fragmentation:** Insights often require cross-document reasoning that simple models cannot perform.
- **Superficial Analysis:** Most existing tools focus purely on sentiment polarity (positive/negative) rather than narrative reasoning or macro-economic implications.
- **Grounding Issues:** Standard LLMs lack real-time market awareness and are prone to hallucinations without a strictly grounded retrieval layer.

1.3 Project Goals & Deliverables

The project will deliver a functional, end-to-end reasoning system designed to produce structured investment "ideas" rather than vague summaries.

- **The System:** A multi-agent pipeline built on LangChain/Index that performs event extraction, macro interpretation, and hypothesis synthesis.
- **The Output:** A series of investment "Data Packets" including asset direction (bullish/bearish), time horizon, a confidence score (0.0 to 1.0), and cited evidence.
- **The Evaluation:** A quantitative report comparing generated signals against 1-week and 1-month future returns using Yahoo Finance data.
- **The Interface:** A Streamlit-based dashboard to visualize the reasoning chain and source citations for each trade idea.

1.4 Alignment with Course Objectives

This project serves as a direct application of core course principles:

- **RAG Implementation:** Using vector databases (FAISS/Chroma) to ground LLM responses in real-world, time-sensitive data.
- **Agentic Workflows:** Utilizing multi-step reasoning to handle complex, non-linear tasks like financial analysis.
- **Systemic Evaluation:** Moving beyond qualitative "vibe checks" to evaluate GenAI outputs using objective, quantitative market metrics.

2. Methodology and Feasibility

2.1 Scope Definition

To maintain technical feasibility, the system will not attempt to track the entire global market. Instead, the scope is defined as follows:

- **Asset Universe:** Limited to the **NASDAQ-100** or **S&P 500 Tech Sector**.
- **Primary Data Feed:** The system will prioritize high-authority financial reporting from **Bloomberg News** and **Yahoo Finance**. Bloomberg provides the "institutional narrative," while Yahoo Finance offers broader retail sentiment and accessible historical price data.
- **Data Granularity:** The system will process daily and weekly signals (medium-frequency), avoiding the infrastructure complexities of high-frequency or intra-day trading.
- **Functional Boundary:** The project focuses exclusively on **hypothesis generation and evaluation**. It will not include execution logic (live trading) or portfolio rebalancing.

2.2 Technical Architecture & Methodology

The system architecture is designed as a modular pipeline:

1. **Data Layer:**
 - * **News Ingestion:** Automated fetching of news via **Yahoo Finance API** and **Bloomberg RSS/Terminal exports** (where accessible).
 - **Metadata Tagging:** Bloomberg articles will be tagged with a "High-Authority" weight in the RAG pipeline to prioritize professional analysis over general web noise.
 - **Market Data:** Historical and real-time price data will be pulled via the **yfinance** library for post-hoc validation.
2. **RAG Layer:** Document chunks will be vectorized and stored in a **FAISS** index. We will implement a "Source-Aware" retrieval strategy that can filter specifically for Bloomberg-sourced context when high-conviction signals are required.

3. **Agent Orchestration:**
- * **The Analyst Agent:** Responsible for document retrieval, specifically cross-referencing **Bloomberg's macro reporting** with company-specific news .
 - **The Strategist Agent:** Interprets sentiment and sector implications .
 - **The Decision Agent:** Synthesizes the output into a structured JSON format .

2.3 Resource Assessment & Availability

All required resources are currently accessible, ensuring no "start-up" lag:

- **Compute/Models:** Development will occur locally using **Python 3.10+**. We will utilize **GPT-4o** via API for the complex reasoning agents and **Llama-3 (local via Ollama)** for preliminary data processing to manage API costs.
- **Data Access:** Financial data is freely available through the **Yahoo Finance API** and open RSS feeds.
- **Software Libraries:** We will rely on established open-source frameworks: **LangChain** for agent logic, **Chromadb/FAISS** for vector storage, and **Streamlit** for the UI.

2.4 Implementation Timeline

The project is structured into six discrete weekly milestones to ensure steady progress:

- **Week 1: Infrastructure.** Setup data scrapers for news feeds and historical price ingestion.
- **Week 2: RAG Pipeline.** Implement chunking strategies and vector database indexing; verify retrieval accuracy.
- **Week 3: Agent Logic.** Develop the internal "Analyst" and "Strategist" prompts and the state-management logic for agent handoffs.
- **Week 4: Synthesis & Output.** Refine the "Decision" agent to produce structured, cited investment hypotheses.
- **Week 5: Back-testing.** Run the generated signals against 1-week and 1-month market returns to calculate signal accuracy and Sharpe-like ratios.
- **Week 6: Deployment.** Finalize the Streamlit dashboard and complete the technical report.

2.5 Risk Management (Challenges & Mitigations)

Identified Challenge	Proposed Mitigation Strategy
Model Hallucination	Use a "strict-grounding" prompt where the agent is penalized if it fails to provide a direct citation from the RAG source.

Information Noise	Implement a pre-filtering agent to deduplicate news and discard "clickbait" articles based on source credibility.
Data Alignment	Ensure all news timestamps are normalized to UTC to prevent "look-ahead bias" during the evaluation phase.
Weak Signal Quality	If raw returns are too noisy, shift evaluation to "Relative Strength"—comparing the stock's performance against its sector benchmark rather than absolute price.

3. Innovation and Relevance

3.1 Originality of Approach

Most current financial AI implementations are limited to "Sentiment Scoring" or "Summarization". This project departs from those conventions by introducing **LLM-driven hypothesis generation**. Instead of merely describing the news, the system is designed to formulate an "investment thesis" that mimics the cognitive process of a human analyst—connecting disparate events (e.g., a supply chain disruption) to a specific market outlook (e.g., bearish signal for a specific ticker).

3.2 Connection to GenAI Research Trends

This project sits at the intersection of three active research frontiers in Generative AI:

- **Agentic Workflows:** Moving away from single-prompt interactions toward "Multi-Agent Systems," where specialized LLM instances collaborate, critique, and refine outputs.
- **Knowledge Grounding via RAG:** Utilizing Retrieval-Augmented Generation to solve the fundamental "hallucination" problem in LLMs, which is a prerequisite for high-stakes domains like finance.
- **Long-Context Reasoning:** Implementing cross-document analysis to identify correlations that single-document models often miss, such as the relationship between macro-economic announcements and specific sector rotations.

3.3 Novel Methodology: The Evaluation Loop

A key innovation of this proposal is the **Quantitative Feedback Loop**. While most GenAI projects are evaluated on "text quality" or "human preference," this methodology anchors LLM performance to objective, real-world market data (e.g., next-day and 1-week returns). By benchmarking the multi-agent system against a random baseline and simple sentiment analysis, the project provides a rigorous test of whether LLM "reasoning" actually adds alpha beyond basic word-association.

3.4 Potential Impact on the Field

The success of this framework suggests a path toward the automation of **High-Level Financial Research**.

- **For Institutional Analysis:** It demonstrates how to scale the "Associate-level" work of scanning thousands of news items to find actionable catalysts.
- **For Retail Investors:** It provides a tool for "Explainable AI," where users don't just get a "Buy" signal, but a transparent reasoning chain with direct citations to the source material.
- **For AI Research:** It contributes to the ongoing study of how multi-agent systems can be used for "Disagreement Detection"—identifying instances where news sentiment is positive but market reaction is negative, potentially uncovering contrarian opportunities.

4. Advanced Framework & Strategic Extensions

While the core objective focuses on signal generation, the architecture is designed to accommodate several "high-alpha" features that move the system toward professional-grade analysis:

4.1 Disagreement & Contrarian Detection

A unique feature of this system is the ability to detect "market-sentiment divergence". This occurs when news sentiment is overwhelmingly positive, yet the asset price reacts negatively . By identifying these anomalies, the system can flag potential contrarian signals or "priced-in" events that standard sentiment models typically miss.

4.2 Cross-Asset & Sector Reasoning

The "Macro Reasoning Agent" is tasked with identifying inter-market correlations. For example, the system will analyze how news regarding energy prices or Federal Reserve interest rate hikes cascades into specific equity sectors like Tech or Transportation . This multi-layered reasoning ensures the system understands that no stock exists in a vacuum.

4.3 The "Memory Module" for Continuous Learning

To improve accuracy over the six-week period, we will implement a lightweight memory module. This allows the system to track the outcome of past hypotheses and "learn" which types of news events consistently lead to successful predictions for specific tickers .

4.4 Explainability & Source Transparency

A critical deliverable is the **Explainability Layer**, which will be integrated into the Streamlit interface. For every generated idea, the system will:

- **Highlight specific evidence:** Extract and display the exact sentences from the retrieved news that drove the conviction.
- **Visualize the Agentic Path:** Show the "chain of thought" from the Event Extraction Agent to the final Decision Agent, ensuring the user can audit the AI's logic.

Summary of the Integrated Project Scope

Field	Description	Example
Asset	The specific ticker being analyzed	NVDA
Signal	Directional bias (Bullish/Bearish/Neutral)	Bullish
Horizon	The expected timeframe for the move	Short-term (5-day)
Confidence	A score (0.0 to 1.0) based on evidence strength	0.72
Reasoning	A concise synthesis of the catalyst	"Earnings beat + AI demand"
Sources	Direct citations to the grounding documents	["Doc1", "Doc2"]